Defining the navigation problem

- One of the classic applications of xKF is navigation.
- Target tracking is the estimation of the state (e.g., position, velocity) of a moving object based on remote observations of the object.
  - Answers the question, “Where is it?”
- Navigation is the estimation of my own state (e.g., position, velocity) using local sensors.
  - Answers the question, “Where am I?”
- Guidance computes a path to a destination; it is a control problem whereas target tracking and navigation are estimation problems.
  - Answers the question, “How do I get there?
- Design criteria for navigation systems include cost, accuracy, physical size, security, reliability (sometimes in harsh environments), and sometimes world-wide availability.

Fusion of relative and absolute measurements

- Most navigation systems combine (“fuse”) data from sensor measurements describing relative changes in position (e.g., velocity, acceleration) with sensor measurements encoding absolute position.
- Using relative measurements only is called “inertial navigation” and suffers due to errors in the initial state estimate, and due to integrating sensor noises and biases.
  - It is like running an xKF open-loop, without measurement feedback.
- Using absolute measurements only is also sub-ideal, especially if the measurement frequency is low and/or if measurement noise is high.
- Combining both types of measurement allows us to take advantage of the benefits of both types and mitigate the disadvantages of both.

Inertial navigation systems (INS)

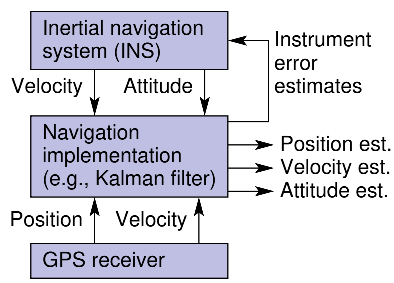
- Inertial navigation systems (INSs) use “dead-reckoning.”
- Such a system is self-contained, nonjammable, nonradiating.
- Its sensors are usually a suite of accelerometers and gyroscopes (“gyros”) which measure linear acceleration and rotational rate (angular velocity).
- To provide 3d navigation with 6 degrees of freedom (6-DOF), three accelerometers are mounted on orthogonal axes and three gyros are positioned to define a 3d coordinate system.
- Gimbaled systems control the orientation of the gyros such that they are isolated from the rest of the vehicle, providing a constant reference frame.
- Strapdown systems mount the gyros and accelerometers to the vehicle body directly; the reference frame is maintained mathematically.
- Gimbal produces better estimates with lower-accuracy sensors, but gimbal control is complex; advances in laser/optical gyros make strapdown systems attractive.

Global positioning system (GPS)

- Different kinds of sensors can provide measurements relating to absolute position.
- In an outdoor environment, GPS is very commonly used.
- 32 satellites (may vary but at least 24 are always operational) orbit the earth with 12h periods, in six orbital planes; at least four can be "seen" at all times anywhere on earth.
- VERY precise time information is maintained by the satellites and constantly tuned for accuracy (accurate to 1ns).
- When a GPS receiver decodes the signal transmitted by a GPS satellite, it knows exactly what time the signal was transmitted and the location of the satellite at that moment.
- By combining time/position information from at least four satellite, the receiver can know its own 3d position and time.

GPS/INS integration

- There are different ways to integrate GPS and INS. One is shown here.
- The primary system is the INS (top of figure), which is constantly operating.
  - It integrates the state equation: $\dot{x}(t) = f(x(t), u(t)) + w(t)$.
- GPS (bottom of fig.) provides position information whenever available (often 1Hz, but perhaps up to 50Hz depending on receiver design).
  - Provides $z(t) = h(x(t)) + v(t)$.
- The navigation algorithm (e.g., xKF) produces a state estimate and instrument-error corrections.
  - Helps estimate sensor biases, which can be subtracted internal to INS even if there is a GPS dropout for a while.



Details on GPS are out of scope

- Models of the GPS signal, error budgets, and 9-DOF models of motion are beyond our scope here.
- I recommend that you take a dedicated course on GPS navigation if that is of interest to you. Confer the references on this slide to get started.[1,2]
- My primary intent in this lesson is to show that xKF is an important tool to solving the outdoor navigation problem.
  - You now understand this tool, so you are ready to initiate some extracurricular study to learn GPS/INS in detail.
  - The rest of this week will focus on the indoor navigation problem instead.

[1] Yaakov Bar-Shalom, X. Rong Li, and Thiagalingam Kirubarajan. Estimation with applications to tracking and navigation: theory algorithms and software. John Wiley & Sons, 2001.

[2] Mohinder S. Grewal and Angus P. Andrews. Kalman filtering: Theory and Practice with MATLAB. John Wiley & Sons, 2014.

Summary

- Navigation is an application of xKF that answers the question, "Where am I?"
- Commonly, noisy relative position updates and noisy absolute position measurements are fused to create a single state estimate.
- An inertial navigation system (INS) integrates the relative-position updates to predict position.
- Often, in an outdoor environment, GPS is used as an absolute position feedback mechanism to update the prediction.
- In an indoor environment, different sensors are used, as you will learn in the next lesson.

Credits

Credits for photos in this lesson

- "GPS" on slide 4: Pixabay license (`https://pixabay.com/en/service/license/`), from `https://pixabay.com/vectors/gps-location-satellite-earth-6327896/`
- Block diagram on side 5 was adapted from: Mohinder S. Grewal and Angus P. Andrews. Kalman filtering: Theory and Practice with MATLAB. John Wiley & Sons, 2014.